



BDM Capstone Project

Project Title

Studying and Enhancing Operational Efficiency at
Mahak Spices Company



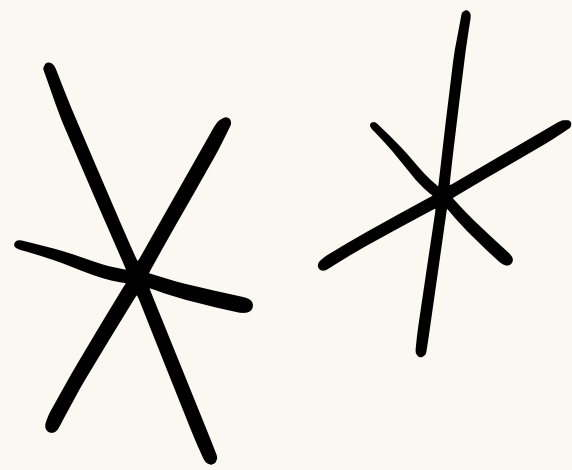
Submitted By :

Name: Shreya Saxena
Roll No: 22f3001013

Overview

- Founded: August 2024 by Akshat & Mahak Jauhari
- Business Type: B2B wholesale supplier of spices
- Location: Subhash Nagar, Shahjahanpur, Uttar Pradesh
- Prime Spot: Close to main trading market, ensuring easy customer access
- Products: Wide variety of fresh, carefully selected spices
- Reputation: Known for quality products and thoughtful planning
- Goal: To become one of the most trusted spice suppliers in the region





OBJECTIVES

1. OVERDEPENDENCE ON FEW SPICES

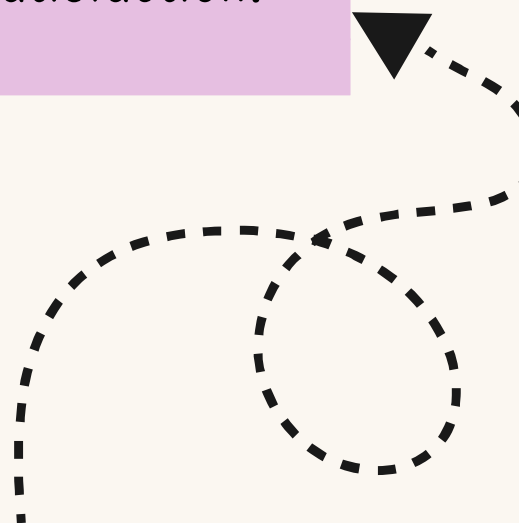
The business relies heavily on few spices leading to poor sales performance for other products. This leads to overstocking of unproductive items.

2. FLUCTUATING TRANSPORTATION COSTS

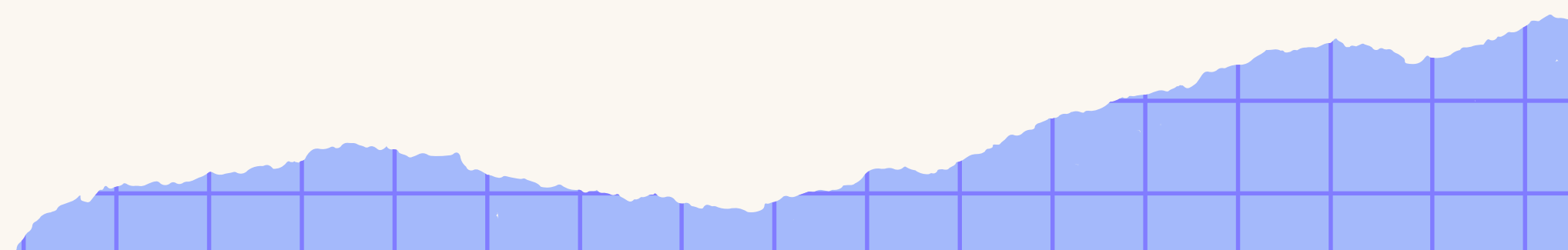
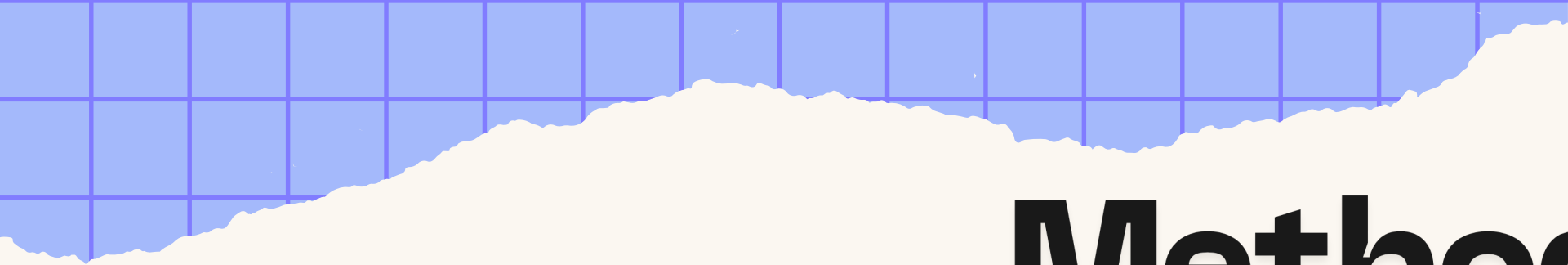
The involvement of third-party service providers causes significant variations in transportation costs. These unpredictable expenses contribute to degradation in the profit margins.

3. SPICE SPOILAGE IN SUMMER MONTHS

High temperature in summers leads to moisture loss in certain spices, resulting in reduced weight and compromised product quality. The spoilage affects both the profitability and customer satisfaction.



Methodology



DATA COLLECTION

- Provided by Mr. Akshat Jauhari owner of Mahak Spices.
- Timeframe: November 2024 to April 2025.

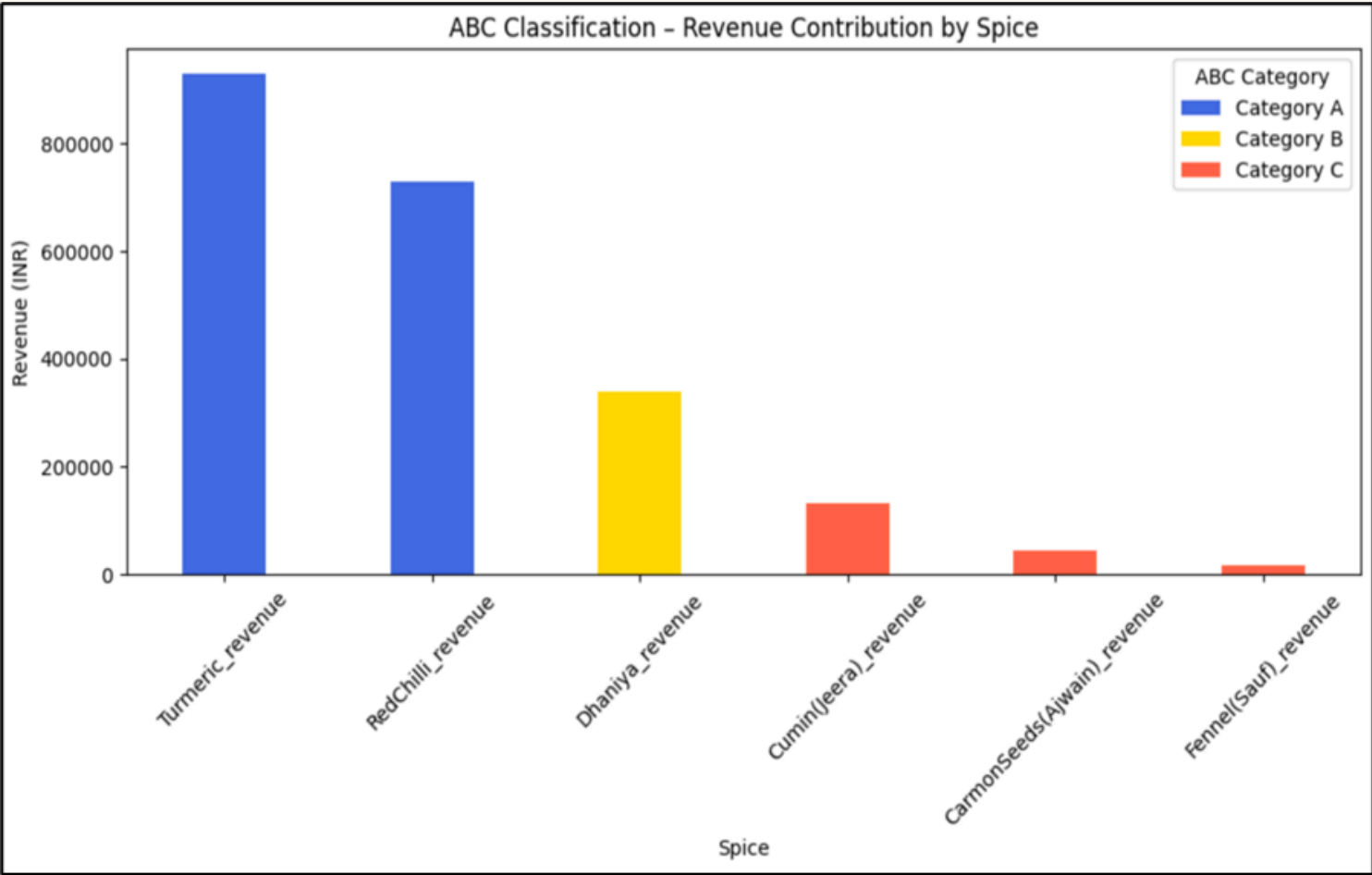
DATA CLEANING AND PREPROCESSING

- Missing values were replaced with zeroes where appropriate.
- Incorrect or inconsistent entries were checked, especially for numeric and date fields.
- Column headers were renamed using consistent formats

DATA ANALYSIS

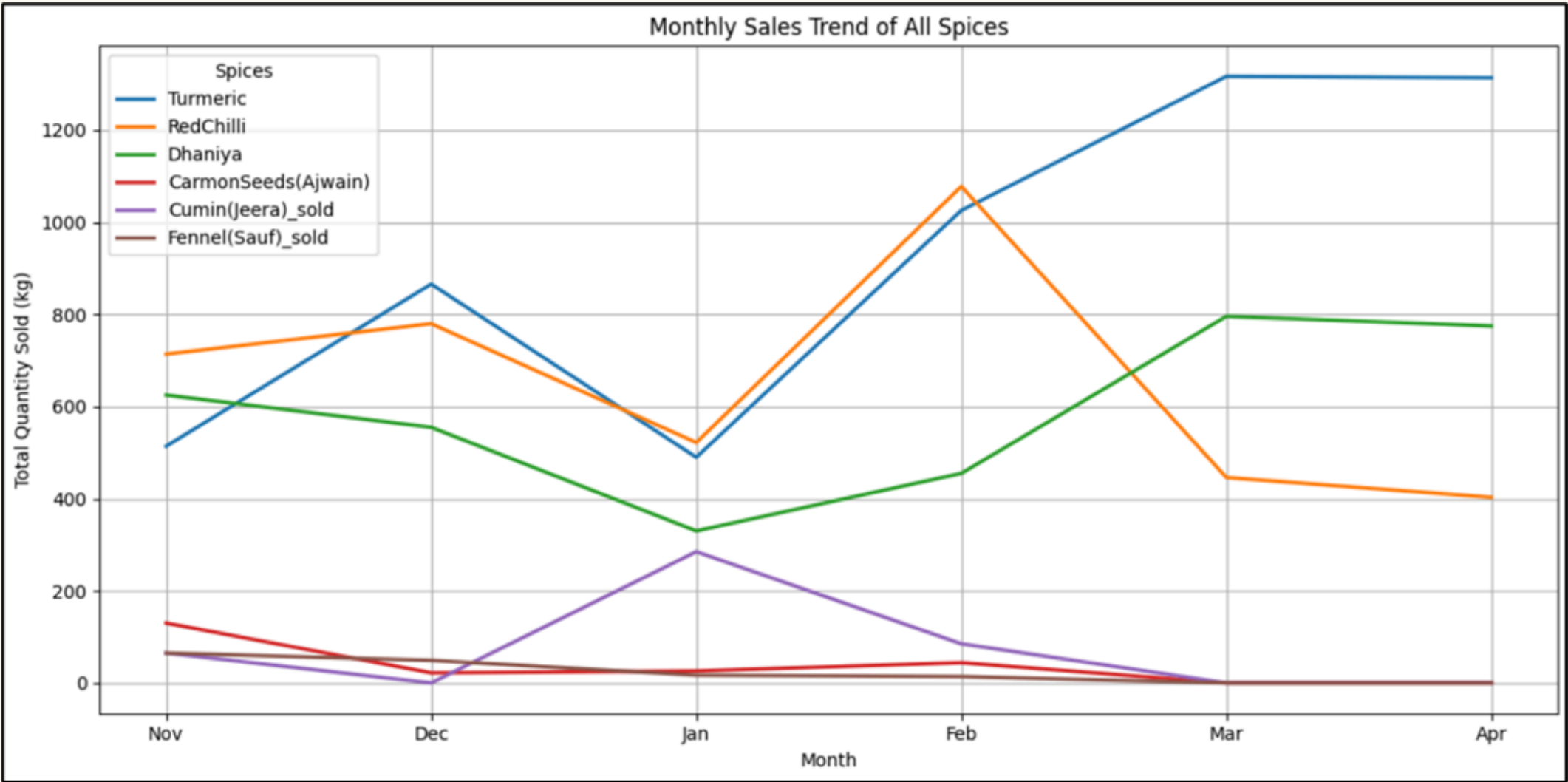
- ABC Analysis.
- Correlation Analysis.
- Trend and Seasonal Analysis
- Profitability and Transport Cost Analysis

TREND OF ALL SPICES



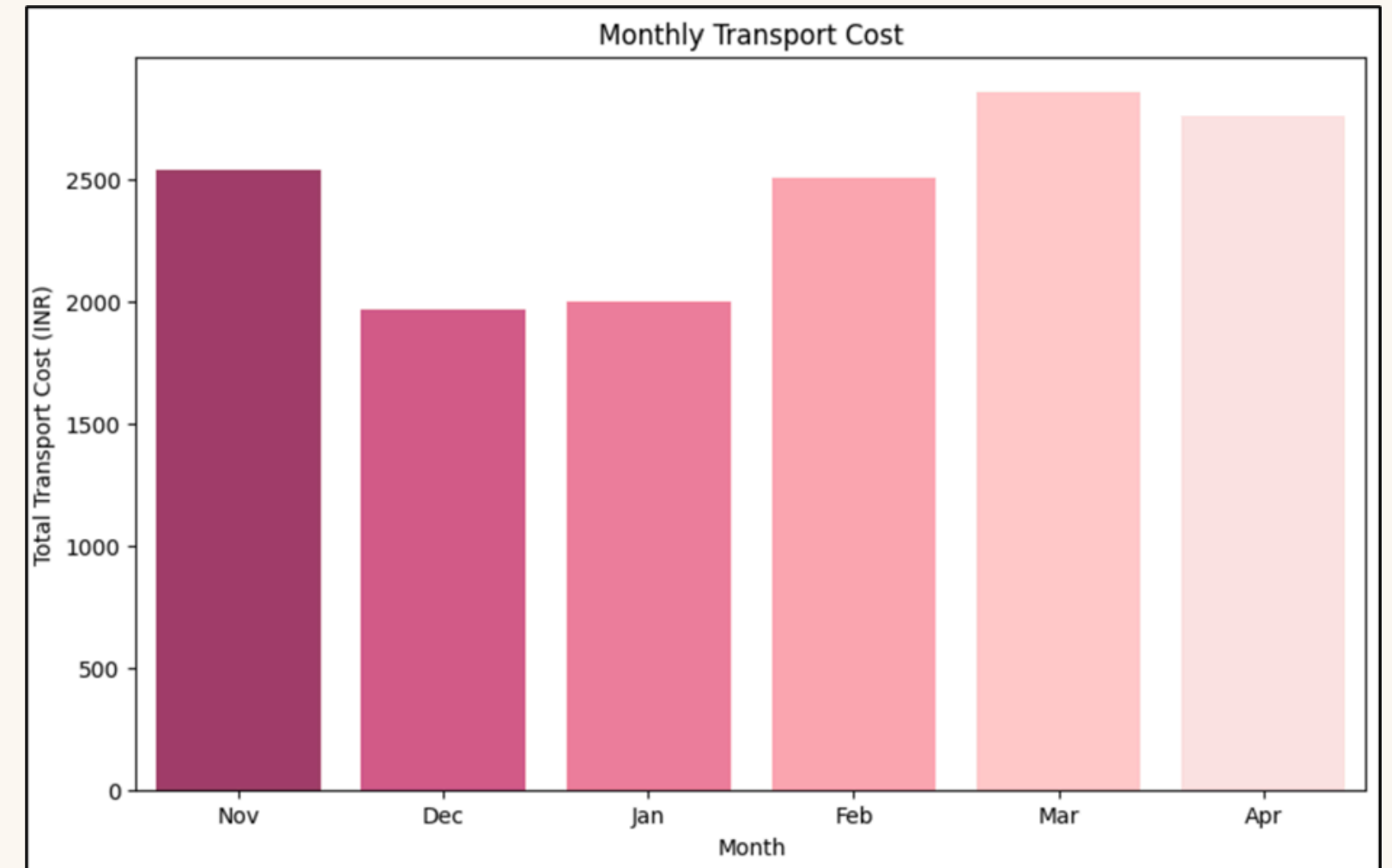
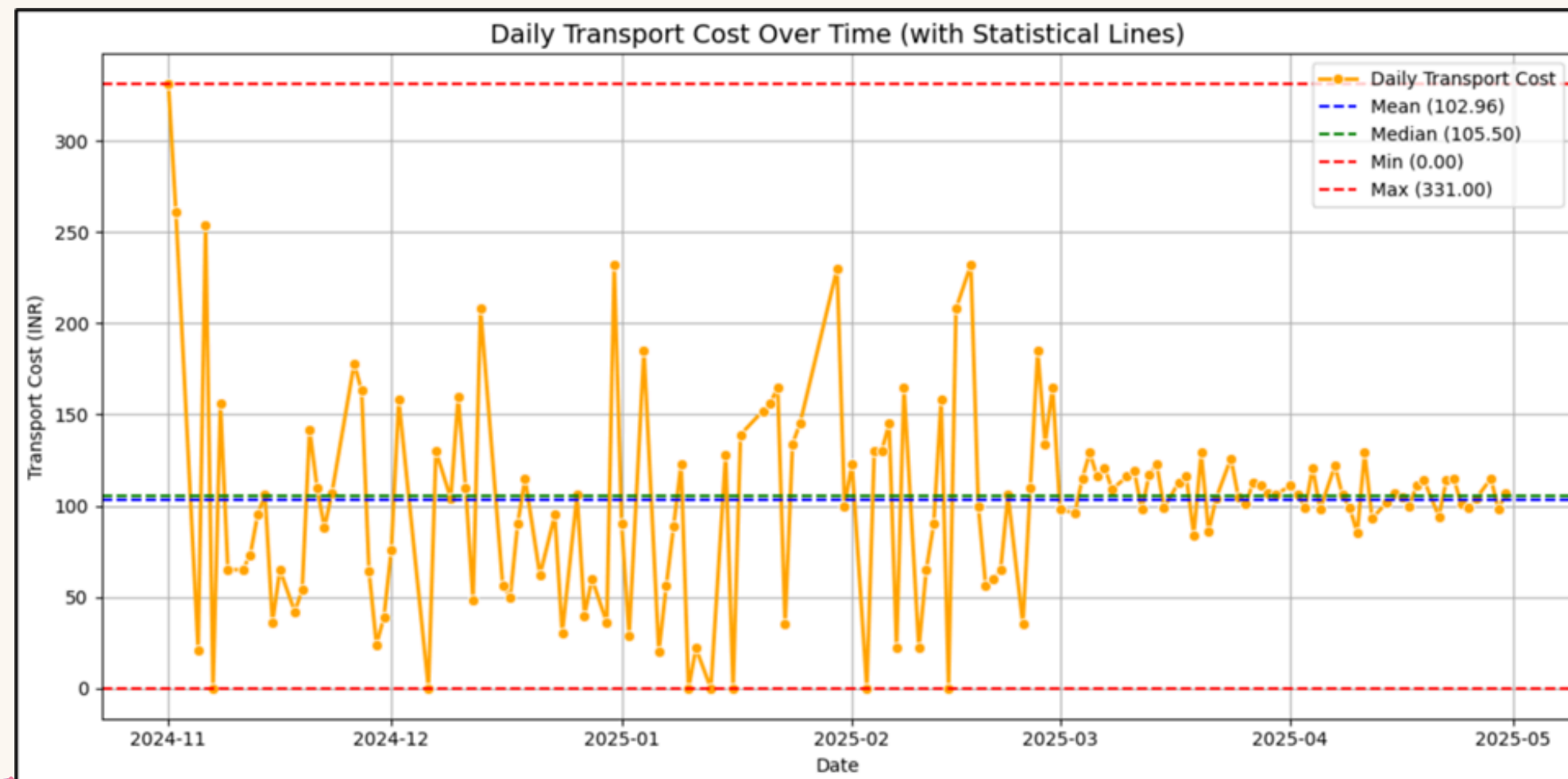
Turmeric and Red Chili are classified as Category A products. Dhaniya is in Category B and Cumin, Carmon Seeds and Fennel are under Category C products

- 1. Turmeric shows consistent, steady growth with high demand
- 2. Red Chilli performs strongly until February but drops sharply afterward, highlighting the need for better perishability management.
- 3. Dhaniya declines early but rises again from February–April, suggesting seasonal or festive demand
- 4. Cumin (Jeera) shows an unusual January sales spike worth investigating, while Ajwain and Fennel (Saunf) remain low-performing



TRANSPORT COST ANALYSIS

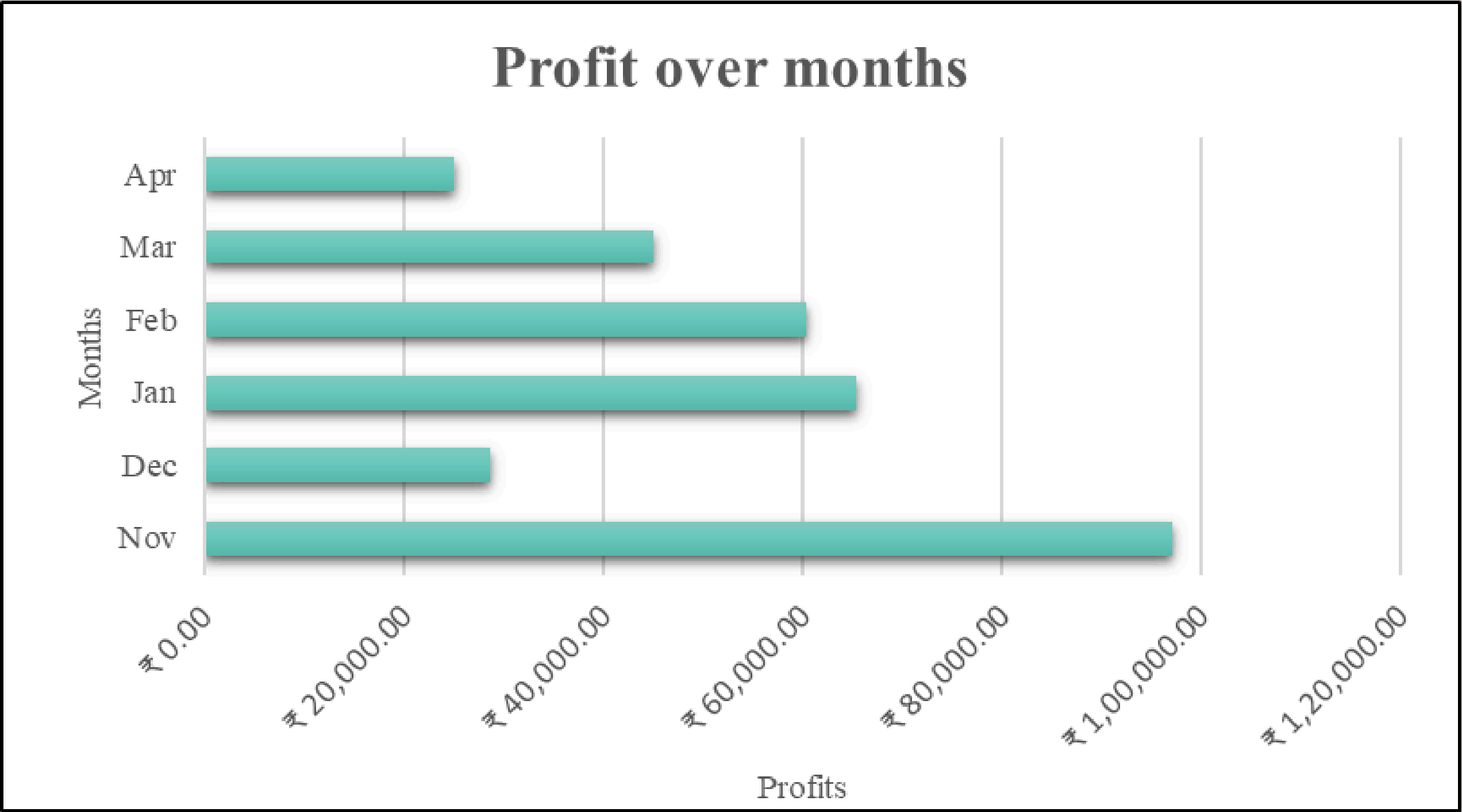
1. Feb-Mar Alignment – Both sales and transport costs rise significantly in these months, showing a direct link between higher demand and increased logistics activity.
2. Transport costs dip in Dec-Jan, which aligns with moderate/declining sales for Dhaniya and consistently low-performing spices, reducing shipping needs.
3. The sharp rise in costs from Feb to Mar mirrors the seasonal surge in spice sales



The line chart indicates huge fluctuations in the daily transport costs.

PROFIT OVER MONTHS

- 1. Highest profit in November despite high transport costs driven by Turmeric & Red Chilli demand.
- 2. Dec Dip – Sharp profit fall due to seasonal slowdown lower sales & transport costs.
- 3. Jan-Feb Recovery – Profits rebound with rising spice sales; higher revenues offset transport costs.
- 4. Mar-Apr Decline – Transport costs peak but profits drop from spoilage & weak demand mix.



KEY FINDINGS AND RESULTS

- Turmeric stands out as the top contributor, followed by Red Chili. Together, Turmeric and Red Chili account for nearly 76% of all revenue.
- March has the highest transport cost, which might suggest a surge in operations or inefficient logistics at that time.
- Red Chili displays solid sales in the initial months (November to February), peaking in February with impressive quantities sold. However, this is followed by a steep decline in March and April, which may be attributed to spoilage due to heat and poor storage conditions during the summer months.
- Cumin, Carmon Seeds and Fennel are classified as Category C spices.
- Turmeric has the highest positive correlation with transport cost. This indicates that whenever Turmeric sales are high, there is a tendency for transportation cost to be high as well. This may be because Turmeric has greater weight or bulk orders than other spices, so logistics become costly.

RECOMMENDATIONS

1. **Diversify Revenue Streams** - Growth of underperforming spices, could be raised with targeted marketing, shelf placement, or combo packaging.
2. **Introduce Combo Offers and Bundling Strategies** - Combine popular spices like turmeric with less popular products
3. **Track Emerging Consumer Demand** – To see if demand is moving toward other products, use sales data and trend analysis.
4. **Implement Long-Term Vendor Contracts** - Collaborating with a regular logistics provider through contracted monthly or per trip rates can help smoothen cost fluctuations
5. **Route Optimization** - Usage of Google Maps, Routific and other digital tools to batch deliveries going to the nearby areas.
6. **Seasonal Inventory Adjustment** – During the summer, reduce the amount of red chili that is ordered or kept on hand. To better match inventory with what can actually be kept and sold during these dangerous times
7. **Invest in Temperature-Controlled Storage** - Use insulated bags or cold storage solutions for transportation, especially during the summer, to lower the possibility of spoiling.
8. **Dynamic Pricing for Near Expiry Stock** - As red chilli approaches its expiration or freshness limit, apply price reductions. Customers feel a sense of urgency as a result, which speeds up stock turnover and helps turn possible losses into profits.

THANK
YOU!

